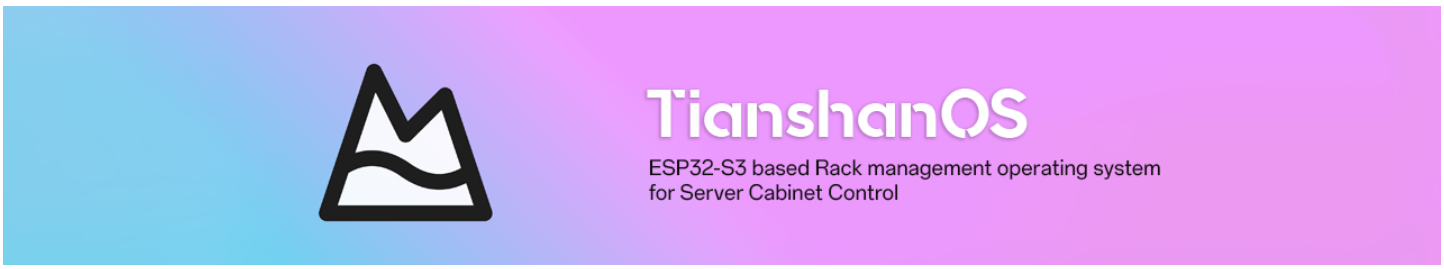
TianshanOS 用户入门指南 / TianshanOS Getting Started Guide



TianshanOS 用户入门指南
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TianshanOS User Getting Started Guide
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TianshanOS 用户入门指南

系统

资源监控

- 资源监控页面用于显示 RM-01 机身内置的**带外管理芯片**的系统资源情况。

- 点击“**服务**”按钮后弹出的模态框中，显示了带外管理系统中各运行服务的健康状况。
- 点击“**详情**”按钮弹出的模态框中，显示了带外管理系统的**内存**和**存储**使用详情，用来查看各类型内存的使用情况和潜在问题。
- 若不清楚各项服务的功能，请勿进行操作；如已确认服务作用且需处理异常，可手动重启相应服务。

系统总览

- 系统总览页面显示 **TianshanOS** 的运行状态与 **RM-01** 的关键信息。
- 在系统总览页面，您可以查看系统信息与电源状态，并执行以下操作：重启 **TianshanOS** 系统、设置关机逻辑（仅适用于配备锂电池的机型）或配置 RM-01 顶部 USB 接口。在执行任何关键操作前，请评估并确认其可能带来的影响。
- USB 按钮用于在**带外管理芯片、推理模组与应用模组**之间切换 RM-01 顶部 USB 接口的连接目标，以适配不同场景或外设配件，默认目标为 **ESP**。误操作可能导致当前设备断连或功能异常。若不清楚该功能或未配备相应配件，请先咨询经销商或运维人员，切勿随意操作。
- 当 **TianshanOS** 出现加载不完整、操作卡顿或其他异常时，可通过“重启”按钮重启带外管理系统以恢复服务。重启过程可能会引起短时中断；如不确定影响范围，请先联系经销商或运维人员。
- **电源保护**按钮用于启用或禁用低电压保护策略：在输入电压异常下降时自动触发关机与恢复流程，以防止设备在低电压下长期运行导致损坏或不稳定。

网络与时间

- 网络与时间集中显示网络连接与当前 IP、设备时间及同步状态与时区，并提供时间同步、时区设置和 OTA 升级入口，便于快速检查与调整。
- **同步时间**按钮用于将当前浏览器的系统时间写入带外管理系统（TianshanOS），用于离线 / 无 NTP、RTC 电池失效导致的设备时间不准或需要快速校时的场景。
- **时区**按钮用于选择预设或输入自定义 POSIX 时区并应用到带外管理系统，用于设置 UTC 偏移与夏令时规则，正确显示和解析本地时间。
- 点击“**OTA 升级**”会打开固件升级页面，为系统固件及 WebUI 提供一站式在线与离线升级管理。
  - OTA 模式：浏览器从服务器下载固件，再通过 API 上传到 ESP32（设备可不连外网）。升级结束后自动重启，页面会尝试检测设备重新上线。
  - 分区管理
    - 运行中分区：显示当前启动分区的固件版本、存放地址与大小；可选择“标记有效”以取消自动回滚机制。

        - 可启动分区：指未在运行的另一分区，若其中已有可启动固件，可点击“回滚到此版本”并重启后从该分区启动以恢复到该固件。
        - 切换/回滚需重启生效，标记有效与回滚操作是确保升级稳定性与快速恢复的关键手段。
    - 手动升级
        - 从 URL 升级：在输入框填入固件地址，可勾选“包含 WebUI”以在固件刷写后继续刷写 `www` ；“跳过验证”可绕过完整性/签名校验（存在较大风险）。适用场景：固件托管于 HTTP/HTTPS 服务器，RM-01 或浏览器需能访问该 URL。
        - 从 SD 卡升级：填写 SD （TF / MicroSD） 卡上的固件路径（如 `/sdcard/firmware.bin` ），勾选“包含 WebUI”时会同时查找并刷写同目录下的 `www.bin` ；适用于设备离线或无外网环境的本地升级。

> ⚠ **风险与注意**
>
> 勾选跳过验证会增加刷写错误或刷入恶意/损坏固件的风险；包含 WebUI 时为两步刷写，整体耗时增加且需确保两者版本兼容。

设备面板

- 整体布局：设备面板由上下两个区域和右上角三个按钮组成。上区域为快捷操作，下区域为数据监控，三个按钮分别为 推理模组电源控制、应用模组电源控制和组件管理。
- 上部区域
    - 快捷操作会展示为可点击的卡片，每个卡片对应一条自动化规则。
    - 来源与配置：卡片来自自动化系统（发行商或经销商的专用界面）中，已启用且勾选了 「**手动触发**」 的规则，发行商或经销商可预先配置这类规则提供给用户。
    - 使用方式：点击卡片即执行对应规则。例如：启动应用、执行脚本、发送命令等。
    - 附加能力（若规则包含后台执行）：
        - 显示进程运行状态（绿点为运行中）
        - 提供「**日志**」和「**停止**」按钮，用于查看日志或终止进程
        - 进程运行中时，再次点击卡片会提示“进程正在运行中，请先停止”
        - 空状态：若没有符合条件的规则，会显示“暂无快捷操作”，并提示在自动化规则中启用「**手动触发**」选项。
- 下部区域
    - 下部区域为用户自定义的数据组件，用于实时显示监控数据。
    - 添加与编辑：点击右上角**组件管理**，在弹窗中：

        - 左侧可设置自动刷新间隔（如 1 秒、5 秒、10 秒或禁用）
        - 可添加新组件（从预设或自定义类型）
        - 可编辑、上 / 下移动、删除已有组件
    - 也可以直接点击面板上的某个组件卡片，进入该组件的编辑界面。
        - 预设组件类型：CPU、内存、硬盘、温度、GPU、功耗、电压、电流、网速、状态、运行时间、日志流等，用于快速添加常用监控。
        - 自定义组件类型：环形进度、仪表盘、温度计、数字、进度条、文本、状态、图标、双值、百分比、日志流等，可绑定任意自动化变量表达式（如 `\${cpu.usage}` 、`\${power.watts}`）。
        - 配置保存：组件配置会保存到设备（SD 卡优先，NVS 备份），换浏览器或设备重启后仍会保留。
        - 空状态：未添加任何组件时，显示“还没有添加数据组件”，引导用户通过组件管理添加。
- 设备面板右上角的两个绿色按钮分别监控着**推理模组**（AGX 运行中）和**应用模组**（LPMU 运行中）的状态；**绿色**代表设备正在运行，**黄色**代表正在获取设备状态（通常是启动中或连接中），**红色**代表已关闭（断电）。用户也可以通过点击来强制切换模组的启动 / 关闭。

> ⚠ **安全提示**
>
> - 在模组运行中强制断电或触发电源切换，可能导致数据、内容不可逆损坏或丢失。
> - 执行前请确认当前工作已保存，或已了解可能后果。

风扇控制

- 整体布局：风扇控制位于系统总览页，与设备面板并排。包含**温度状态栏**、**风扇卡片网格**及右上角**刷新**、**曲线**两个按钮。
- 右上角按钮
    - **刷新**按钮：用于重新获取当前风扇状态（模式、转速、RPM、温度等），确保显示为最新数据。
    - **曲线**按钮：打开风扇曲线管理弹窗，用于配置温度-转速曲线及相关参数。
- 温度状态栏，顶部横排展示三类信息
    - 有效温度：可用于控速的温度值（°C），来自绑定的温度变量；若未绑定或无效，会以警告色显示
    - 目标转速：在曲线/自动模式下，按当前温度计算得到的目标占空比（%）；手动模式下为实际当前转速

- 测试温度：用于调试曲线逻辑的手动温度输入

> ⚙️ 测试温度功能
>
> - 在输入框中填入 0–100 的整数（单位 °C），点击「测试」即可将温度源切换为该固定值，风扇将按此温度根据曲线计算目标转速。
> - 适用于在真实温度变化不明显时，快速验证不同温度点下的转速表现。
> - 点击「清除测试」会恢复使用实际温度变量，不再使用手动温度。
>
> **注意**：测试温度生效期间，风扇控速不再依赖真实传感器，仅用于调试，验证完应尽快清除。

- 风扇卡片
  - 卡片标题与模式
    - 左上角：风扇编号，如「**风扇 0**」。
    - 右上角：当前模式徽章
      - **关闭**：风扇停转
      - **手动**：由用户指定固定转速
      - **自动**：使用系统默认控速策略
      - **曲线**：使用自定义温度-转速曲线
  - 转速显示
    - 大号数字为占空比（0–100%）。
    - 曲线/自动模式：显示目标转速。
    - 手动模式：显示当前设定转速。
    - 若风扇支持 RPM 反馈，下方会显示实际转速（RPM）。
  - 模式切换：四枚 Tab 按钮：**关闭** | **手动** | **自动** | **曲线**，点击即可切换该风扇的运行模式。
  - 转速调节滑块
    - 仅在手动模式下可用；在关闭、自动、曲线模式下为灰色不可用。
    - 范围 0–100%，拖动滑块或输入数值即可设定风扇转速。

> ℹ️ **提示**：「切换到手动模式后可调节」说明需先切到手动模式才能使用滑块。

- 曲线管理弹窗通过点击**曲线**按钮打开，可配置温度-转速曲线及其相关参数。
  - 风扇选择：顶部下拉框选择要配置的风扇（如 0、1 等）。

- 温度源（绑定变量）
  - 作用：曲线模式需根据温度计算转速，温度来源为绑定的自动化变量（如 `agx.cpu_temp`）。
  - 操作：在变量下拉框中选择一个浮点型变量，点击「**绑定**」；可随时「**解绑**」或更换变量。
  - 状态：显示当前绑定变量及其实时温度值；未绑定时会显示“未绑定”。
  - **提示**：可选择类似 `agx.cpu_temp` 的 CPU 温度变量作为温度源。
- 温度-转速曲线
  - 由若干 （温度， 转速） 点组成，每个点表示「当温度达到 X°C 时，目标转速为 Y%」。
  - 添加点：点击「**添加点**」，在末尾新增曲线点（最多 10 个）。
  - 删除点：每个点旁的删除按钮可移除该点（至少需保留 2 个点）。
  - 编辑：在输入框中修改温度（-20–120°C）和转速（0–100%）。
  - **规则**：温度低于最低点时使用最低转速；高于最高点时使用最高转速；中间按曲线线性插值。

- 曲线预览：画布实时绘制当前曲线的温度-转速关系，便于检查配置是否合理。
- 高级参数

参数	含义	范围 / 说明
最小占空比	曲线计算出的目标转速下限	0–100%，通常 20% 以上可避免停转
最大占空比	曲线计算出的目标转速上限	0–100%，通常 100% 表示全速
温度迟滞	温度上升/下降时的切换阈值差	0–20°C，用于减少在临界温度附近频繁调速
最小间隔	两次调速之间的最小时间间隔	500–30000 ms，用于避免过于频繁调

- 底部操作
  - 取消：关闭弹窗，不保存修改。
  - 导入配置：从本地或 SD （ TF / MicroSD ） 卡加载已有曲线配置。
  - 导出配置：将当前曲线配置导出到本地及 SD （ TF / MicroSD ） 卡。
  - 应用曲线：将当前配置写入设备并生效。

> ℹ️ **注意事项**
>
> - 曲线模式需先绑定温度变量，否则无法正确按温度控速。
> - 最小占空比不宜设为 0，否则低温时可能长时间停转。
> - 测试温度只用于调试，测试结束后务必清除测试以恢复正常控速。

LED 控制板块

- 整体布局：LED 控制位于系统总览页，在风扇控制下方，由标题栏按钮和设备卡片网格组成，每个物理 LED 设备对应一张卡片。
- 标题栏按钮

按钮	功能
刷新	重新获取所有 LED 设备列表及当前状态
色彩校正	打开“全局色彩校正”弹窗，仅当存在 **Matrix 矩阵屏** 设备时显示
全部关闭	关闭所有 LED 设备，清除显示内容

- 设备类型：系统支持三种 LED 设备，由后端 `led.list` 返回，每类对应一张卡片。

设备名	说明	典型用途
Touch	触摸指示灯（1 颗 WS2812）	状态指示
Board	主板状态灯带（28 颗 WS2812）	状态灯带
Matrix	LED 矩阵屏（16×16）	图像、文本、QR 码、滤镜等

- 设备卡片结构
  - 状态徽章：当前显示状态
    - **已关闭**：设备已关
    - **常亮**：纯色显示，无动画
    - **动画名**：正在播放程序动画
  - 停止动画：当前有动画时可点击，停止当前动画
- 控制区域
  - 亮度滑块
    - 范围 0–255
    - 拖动即生效，会立即下发到设备
    - 右侧显示当前亮度数值
  - 颜色选择
    - 自定义颜色：点击色块打开系统取色器，选择后立即填充到设备
    - 快捷色块：红、橙、黄、绿、青、蓝、紫、白、粉、灰等，点击即可填充；填充颜色会自动开启设备并进入常亮状态。
- 底部操作栏

按钮	功能
电源（灯泡图标）	开关设备：关→点一下为白光常亮；开→点一下为关闭
更多动画（播放图标）	打开“程序动画”弹窗，选择并启动动画
图像/QR码	仅 Matrix 显示，打开图像/QR 码弹窗
文本	仅 Matrix 显示，打开文本显示弹窗
滤镜效果	仅 Matrix 显示，打开后处理滤镜弹窗
保存	将当前显示状态保存为**开机配置**，重启后恢复

- 程序动画弹窗通过点击**更多动画**打开，可切换并配置动画效果。
  - 启用开关：勾选/取消勾选等同于电源开关
  - 动画列表：按钮网格，点击选中并预览，部分动画支持颜色参数（如 breathing、solid、rain）
  - 设置：选中后显示
  - 速度：1–100 滑块，控制动画速度
  - 颜色：仅部分动画显示，用于选择主色
  - 操作：重置、应用（启动选中动画并关闭弹窗）
  - **说明**：动画种类由后端 `led.effect.list` 提供，不同设备可能不同。
- 图像 / QR 码弹窗（仅 Matrix）
  - 图像：
    - 路径输入框：填写图像路径（如 `/sdcard/images/xxx.png`）
    - 浏览：打开文件选择器，从 SD 卡选图
    - 居中显示：勾选则图像居中
    - 显示图像：将当前路径图像显示到 Matrix
  - QR 码：
    - 内容输入：输入文本或 URL 生成 QR 码
    - 纠错级别：L / M / Q / H
    - 前景色：QR 码颜色
    - 背景图：可选背景图路径

    生成 QR 码：生成并显示到 Matrix
- 文本显示弹窗（仅 Matrix）
  - 文本内容：输入要显示的字符串
  - 字体：从系统 / SD 卡字体列表中选择
  - 对齐：左 / 中 / 右

- 颜色：文本颜色
- 位置：X、Y 坐标，或勾选“自动位置”
- 滚动：无 / 左 / 右 / 上 / 下，以及速度、是否循环
- 显示：应用并显示文本
- 停止：停止文本显示

- 后处理滤镜弹窗（仅 Matrix）：在已有内容上叠加实时滤镜效果。
  - 滤镜列表：脉冲、呼吸、闪烁、波浪、扫描线、故障艺术、彩虹、闪耀、等离子体、怀旧、色阶分离、对比度、反色、灰度等。
  - 操作流程：选择滤镜 → 可选调整参数（速度、强度、波长等）→ 点击「**应用**」→ 点击「**停止**」可停止滤镜。
- 全局色彩校正弹窗：仅当存在 Matrix 设备时，标题栏显示「**色彩校正**」按钮。用于校正 Matrix 的显示色彩。

参数	作用	范围
白平衡	分别调节 R/G/B 通道，修正色温	各通道 0–2.0，默认 1.0
Gamma	中间调对比度	0.1–4.0，默认 1.0
亮度	整体亮度	0–2.0，默认 1.0
饱和度	色彩饱和度	0–2.0，0 为灰度

> ℹ️ **注意事项**
>
> - 保存配置会将当前显示内容写入开机配置，重启后自动恢复。
> - 色彩校正作用于 Matrix 的全局输出，调整会影响所有显示内容。
> - 若未检测到 LED 设备，请检查 LED 服务状态和 GPIO 配置。

网络

整体布局

网络页面通过侧边栏「**网络**」进入，由网络状态概览、接口配置、网络服务三个主区域，以及若干可展开的辅助面板和弹窗组成。

网络状态概览

页面顶部的横排展示当前三个网络接口的简要状态：

接口	显示内容	状态含义
以太网	链路状态（已连接/未连接）、IP 地址	绿=已连接，灰=未连接
WiFi 客户端	连接状态（已连接/未连接）、IP 地址	绿=已连接，灰=未连接
WiFi 热点	运行状态（运行中/未启用）、接入设备数	绿=运行中，灰=未启用

接口配置面板

通过「**以太网**」「**WiFi**」两个 Tab 切换。

- 以太网，展示以太网接口的只读状态，不提供修改入口：

项	含义
链路状态	已连接 / 未连接
IP 地址	本机 IP
子网掩码	子网掩码
网关	默认网关
DNS	DNS 服务器
MAC 地址	网卡物理地址

> ℹ️ **说明**：以太网通常由上级网络分配地址（如 DHCP），或通过其它配置方式设置。

- WiFi
  - WiFi 模式选择：
    - 下拉选择：`关闭 / 站点（STA） / 热点（AP） / STA+AP`
    - 选择后立即切换模式并刷新页面
  - 站点连接区域（STA 或 APSTA 模式时显示）：

项	含义
状态	已连接 / 未连接
SSID	当前连接的 WiFi 名称
IP	本机 IP
信号	信号强度 dBm 及简要指示条

- 操作按钮
  - 扫描：打开 WiFi 扫描面板，列出可用网络
  - 断开：断开当前 WiFi（仅在已连接时显示）
- 热点区域（AP 或 APSTA 模式时显示）：

项	含义
状态	运行中 / 未启用
SSID	热点名称
IP	热点 IP（如 192.168.4.1）
接入数	当前连接的设备数量

- 操作按钮：
  - 配置：打开热点配置弹窗
  - 设备：打开热点接入设备列表

WiFi 扫描面板

点击「**扫描**」后展开，显示可用 WiFi 列表。

- 每个网络显示：SSID、信号强度 dBm、信道、认证方式
- 点击某条网络即弹出密码输入框，输入后发起连接（开放网络可留空）
- 刷新：重新扫描
- 关闭：收起面板

> ℹ️ **说明**：在 STA 或 APSTA 模式下才能扫描；若模式为 off 或 ap，会出现“需要切换到 STA 或 APSTA 模式”的提示。

热点接入设备面板

点击「**设备**」后展开，列出当前接入热点的设备。

- 每个设备显示：MAC 地址、信号强度 dBm
- 关闭：收起面板

DHCP 客户端面板

点击「客户端」后展开，显示 DHCP 分配情况。

- 接口选择：选择「**WiFi AP**」或「**Ethernet**」
- 列表中显示：IP、MAC、主机名（如有）、类型（静态/动态）
- 刷新：重新获取客户端列表
- 关闭：收起面板

网络服务面板

- 主机名
  - 当前主机名（如 TianshanOS）
  - 输入新主机名 → 点击「设置」→ 更新主机名并刷新页面
- DHCP 服务器
  - 状态徽章：显示运行中的 DHCP 接口数量（如 2/2）或总状态
  - 接口列表：各接口的运行状态和租约数量
  - 客户端：打开 DHCP 客户端面板，查看分配情况
- NAT 网关
  - 状态徽章：运行 / 停止
  - 接口状态：WiFi、以太网 是否可用（用于 NAT）
  - 启用/禁用：切换 NAT 开关
  - 保存：保存 NAT 配置

> ℹ️ **说明**：NAT 用于在 WiFi 和以太网之间共享上网。启用需同时满足：
>
> - WiFi 已连接，且
> - 以太网已连接
>
> 否则按钮会被禁用。

WiFi 热点配置弹窗

点击「配置」打开，用于设置 AP 或 APSTA 模式下的热点参数。

Z	含义	说明
SSID	热点名称	必填
密码	无线密码	留空为开放网络；若填写，至少 8 位
信道	工作信道	2001/6/11
隐藏 SSID	是否隐藏热点名称	勾选后需手动输入 SSID 才能连接

使用流程示例

- 连接 WiFi：在 WiFi Tab 选择 STA 或 APSTA → 点击「**扫描**」→ 选择网络并输入密码 → 连接
- 断开 WiFi：在已连接状态下点击「**断开**」
- 开启热点：选择 AP 或 APSTA → 点击「**配置**」→ 填写 SSID、密码等 → 应用
- 查看热点接入设备：点击「**设备**」→ 展开接入设备列表
- 启用 NAT：确保 WiFi 和以太网都已连接 → 点击「**启用**」
- 修改主机名：在服务区输入新主机名 → 点击「**设置**」
- 查看 DHCP 客户端：点击「**客户端**」→ 选择接口 → 查看分配列表

ℹ️ 注意事项

- 以太网配置为只读展示，修改需通过上级 DHCP 或其它方式
- 扫描和连接 WiFi 需要 STA 或 APSTA 模式
- 热点密码至少 8 位，或留空作为开放热点
- 启用 NAT 前需确认 WiFi 和以太网均已正常连接

文件

整体布局

文件页面用于浏览和管理**带外管理系统（TianshanOS）**存储上的文件和目录。主要包含**工具栏**、**存储切换**、**导航**、**文件列表**、**卸载**等区域和按钮。

工具栏

按钮	功能
上传文件	打开上传弹窗，将本地文件上传到当前目录
新建文件夹	打开新建文件夹弹窗，在当前目录创建子文件夹
刷新	重新加载当前目录列表

存储切换

两个标签用于切换存储位置：

标签	路径	说明
SD 卡	/sdcard	外置 SD 卡存储
SPIFFS	/spiffs	板载 Flash 存储

点击对应标签会跳转到该存储的根目录。

SD 卡控制：

- 已挂载：显示「卸载 SD」按钮，可卸载 SD 卡
- 未挂载：显示「挂载 SD」按钮，可挂载 SD 卡
- 当前为 SD 卡路径且 SD 未挂载时，列表区会显示「SD 卡未挂载」和「挂载 SD 卡」按钮

存储状态：底部显示 SD、SPIFFS 的挂载状态（已挂载 / 未挂载）。

文件列表

表格形式展示当前目录内容，列包括：

列	说明
复选框	选择/取消选择该项，表头可「全选」
名称	文件或文件夹名称；点击文件夹名称进入该目录
大小	文件大小（B/KB/MB/GB），目录显示为 -
操作	下载（仅文件）、重命名、删除

批量操作工具栏

勾选至少一项后，在文件列表上方会出现：

按钮	功能
批量下载	下载所选文件（不包含所选文件夹）
批量删除	删除所选文件和文件夹（需确认）
取消选择	清除当前选择

提示：批量下载时，文件夹会被自动忽略；若只选了文件夹，会提示「选中的项目中没有可下载的文件」。

上传

点击「上传文件」后打开。使用方式：

- 点击上传区域选择文件，或拖拽文件到上传区域
- 支持多选
- 选中的文件会列在上传列表中，可逐一移除

> ℹ️
> - 目标路径：上传到当前浏览目录，文件名不变。
> - 上传限制：后端仅接受写入 /sdcard，上传到 SPIFFS 会失败。
> - 配置包 （。tscfg）：上传 。tscfg 文件时会自动校验签名，通过后可选择是否立即应用。

> ℹ️ **注意事项**
> - 上传路径：仅支持上传到 /sdcard，不能写入 /spiffs。
> - 删除不可恢复：删除后无法撤销，删除前请确认。
> - SD 卡卸载：卸载后无法访问 SD 上的文件；若当前在 SD 路径，卸载后会自动切换到 /spiffs。
> - 批量下载：只下载选中的文件，所选文件夹会被自动忽略。

安全

- 此章节为「**安全**」页面简介和操作摘要，详细操作步骤参阅 **安全页面完整操作指南（admin 用户）**
- 安全页面是 TianshanOS 的安全管理中心，集中管理 SSH 密钥、远程主机、HTTPS 证书与配置包。需 **admin** 权限，URL 路由为 `#/security`。

概述与准备

- 如何进入安全页面
  a. 在浏览器中访问设备的 WebUI（如 `http://192.168.4.1`）
  b. 使用 admin 账户登录
  c. 在顶部导航栏点击**「安全」**链接
  d. 进入安全页面（`#/security`）
- 页面结构

  页面自上而下分为五个区块：

区块	功能
密钥管理	生成、导出、部署、撤销、删除 SSH 密钥
已部署主机	已部署公钥的远程服务器列表及操作
已知主机指纹	SSH 服务器指纹，防中间人攻击
HTTPS 证书	设备 mTLS 密钥与证书管理
配置包 (Config Pack)	设备证书导出、配置包导入/导出与列表

密钥管理

- 生成 SSH 密钥
  - 操作步骤
    i. 在「密钥管理」区块点击**「生成新密钥」**按钮
    ii. 在弹出的弹窗中填写表单
  - 表单字段

字段	必填	说明	填写示例
密钥 ID	是	唯一标识，字母、数字、下划线	default、agx_ke
密钥类型	是	加密算法	RSA 2048-bit（推荐）、RSA 4096-bit、EC
备注	否	用途说明	用于连接 AGX Or
别名	否	替代 ID 显示，启用「隐藏密钥」时建议填	AGX 生产环境
允许导出私钥	否	勾选后可导出私钥备份	仅在需要时勾选
隐藏密钥 ID	否	低权限用户无法看到真实 ID	多人共用设备时勾

- 密钥类型说明

类型	生成时间	注意
RSA 2048-bit	5-10 秒	**推荐**，SSH 公钥认证支持
RSA 4096-bit	30-60 秒	高安全，SSH 支持
ECDSA P-256	3-5 秒	**当前不支持 SSH 公钥认证**
ECDSA P-384	3-5 秒	**当前不支持 SSH 公钥认证**

i. 点击**「生成」**按钮，等待完成（RSA 4096 约 30-60 秒，请勿关闭页面）

ii. 弹窗提示成功后关闭，新密钥出现在表格中

> ℹ️ **注意事项**：
>
> - 密钥 ID 不可重复，已存在会报错
> - 生成过程中勿刷新或关闭页面
> - 密钥存储在 NVS 加密分区，重启不丢失

- 部署公钥到远程服务器
  - 前提条件
    - 已生成至少一个 SSH 密钥
    - 知道远程服务器 IP/主机名、用户名、密码
    - 设备与服务器网络连通
  - 操作步骤

    i. 在「**密钥管理**」表格中找到要部署的密钥

    ii. 点击该行的**「部署」**按钮

    iii. 在弹出的「部署公钥到远程服务器」弹窗中填写：

字段	必填	说明	示例
目标主机	是	IP 或主机名	192.168.55.100, agx.local
用户名	是	远程登录用户	root, nvidia
端口	否	SSH 端口，默认 22	22, 2222
认证密码	是	SSH 登录密码，首次部署必需	—

iv. 点击**「开始部署」**

v. 弹窗显示部署过程：连接 → 验证主机密钥（首次需信任）→ 上传公钥 → 验证

vi. 成功后主机会自动加入「已部署主机」列表

- 常见错误

错误	可能原因	处理
连接超时	网络不通、防火墙、SSH 未启动	检查网络、确认 sshd 运行、检查防火墙
密码错误	用户名或密码错误	核对用户名、密码及大小写
权限被拒绝	SSH 不允许密码登录	服务器 sshd_config 中启用 PasswordAuthentication yes
主机指纹验证失败	首次连接需信任	见「已知主机指纹」

- 撤销公钥
  - 从密钥管理撤销
    i. 在「**密钥管理**」表格中找到要撤销的密钥
    ii. 点击**「撤销」** 按钮
    iii. 在弹窗中填写：目标主机、用户名、端口（默认 22）、认证密码
    iv. 点击**「撤销公钥」**，确认后从远程 `authorized_keys` 中移除公钥，本地记录随之删除
  - 从已部署主机撤销
    i. 在「**已部署主机**」表格中找到目标主机
    ii. 2. 点击**「撤销」**按钮
    iii. 输入服务器密码并确认
    iv. 系统删除远程公钥并移除本地记录
  - 仅移除本地记录

    > 若服务器已停用或公钥已手动删除，只需清理本地：

    i. 1. 在「**已部署主机**」中点击**「移除」**
    ii. 2. 确认后仅删除本地记录，不修改远程 `authorized_keys`
- 导出密钥

- 导出公钥
  i. 在「**密钥管理**」表格中找到密钥
  ii. 点击**「Pub」** 按钮
  iii. 在弹窗中复制公钥或下载 `.pub` 文件
  iv. 可将公钥粘贴到远程 `~/.ssh/authorized_keys`
- 导出私钥

  > **前提**：生成密钥时勾选了「**允许导出私钥**」。

  i. 点击**「Priv」** 按钮
  ii. 阅读安全警告并确认
  iv. 复制私钥或下载 `.pem` 文件

  ℹ️ **注意**：私钥泄露可导致未授权访问，勿通过邮件或聊天发送

- 删除密钥
  a. 在「密钥管理」中找到要删除的密钥
  b. 点击**「删除」** 按钮
  c. 确认删除（**不可恢复**）
  d. 密钥从 NVS 永久删除；已部署主机记录仍保留，需手动撤销或移除

已部署主机

- 导入主机配置

  > 适用于：公钥通过终端或手动部署，主机未自动出现在列表中。

  a. 在「已部署主机」区块点击**「导入主机」**
  b. 选择 `.tscfg` 配置包文件
  c. 系统验证签名并显示配置 ID、签名者等
  d. 勾选「覆盖已存在的配置」（可选）
  e. 点击**「确认导入」**
  f. 导入后刷新，主机出现在列表中
- 导出主机配置
  a. 在「已部署主机」表格中找到主机
  b. 2. 点击**「导出」**按钮

c. 可选：粘贴目标设备证书 PEM，用于发给其他设备

d. 点击「导出」，下载 `.tscfg` 文件

- 测试 SSH 连接

  a. 在「已部署主机」中找到目标主机

  b. 2. 点击 ****「测试」**** 按钮

  c. 系统使用部署的密钥连接并执行简单命令

  d. 查看结果：成功显示用户名，失败显示错误信息

- 主机列表字段

列	说明
主机 ID	格式 密钥ID@主机地址
地址	IP 或主机名
端口	SSH 端口
用户名	登录用户
部署密钥	使用的密钥 ID

已知主机指纹

> SSH 连接时自动记录的服务器指纹，用于防止中间人攻击。

- 首次连接新服务器

  首次连接时系统会提示信任主机：

  a. 显示主机地址、密钥类型、指纹 (SHA256)

  b. 选择**yes**：信任并保存指纹，继续连接

  c. 选择**no**：中止连接，不保存

  信任后，指纹出现在「**已知主机指纹**」列表，后续连接自动验证。

- 主机指纹不匹配

  若服务器指纹与保存的不一致，会弹出红色警告。

  可能原因：中间人攻击、服务器重装、密钥更换、IP 重新分配。

- 查看与删除指纹

  - 查看完整指纹

    i. 在「已知主机指纹」表格中找到主机

    ii. 2. 点击「**View**」按钮

iii. 弹窗显示完整 SHA256 指纹

- 删除指纹
  i. 在表格中找到要删除的主机
  ii. 点击**「Delete」**
  iii. 确认删除
  iv. 下次连接该服务器会再次提示信任指纹

> ℹ️ 提示：服务器重装前需先删除旧指纹，否则会持续提示不匹配。

HTTPS 证书

- 证书状态说明

状态	说明
未生成密钥	需先生成密钥对
已生成密钥，未安装证书	需生成 CSR 并安装 CA 签发的证书
已激活	证书已安装且有效
即将过期	证书 30 天内过期，建议续期
已过期	证书过期，mTLS 不可用

- 完整证书申请流程
  - 步骤 1：生成密钥对
    i. 点击**「生成密钥对」**
    ii. 弹窗中直接点击**「生成」**（ECDSA P-256，设备端生成，不导出）
    iii. 若已有密钥，会提示覆盖，确认后继续
  - 步骤 2：生成 CSR
    i. 点击**「生成 CSR」**
    ii. 在弹窗中填写（可选）：

字段	说明	示例
设备 ID (CN)	设备唯一标识	TIANSHAN-RM01-0001
组织 (O)	组织名称	HiddenPeak Labs
部门 (OU)	部门	Device

    iii. 点击**「生成 CSR」**

    iv. 复制 CSR 内容，提交给 CA 服务器签发
  - 步骤 3：安装设备证书
    i. 在 CA 服务器用 CSR 签发证书
    ii. 点击**「安装证书」**
    iii. 将签发的 PEM 证书粘贴到文本域
    iv. 点击**「安装」**
  - 步骤 4：安装 CA 证书链（可选）
    i. 1. 点击**「安装 CA」**
    ii. 粘贴根证书和中间证书（PEM 格式，可拼接多个）
    iii. 3. 点击**「安装」**
  - 步骤 5：查看证书
    i. 点击**「查看证书」**
    ii. 弹窗显示证书 PEM，可复制到剪贴板
- 删除凭证
  a. 点击**「删除凭证」**
  b. 确认后删除私钥、设备证书、CA 链
  c. 操作不可撤销

配置包 (Config Pack)

用于安全加密、签名配置，在设备间分发。

- 导出设备证书
  a. 点击**「导出设备证书」**
  b. 弹窗显示：证书指纹 (SHA256)、证书 CN、证书 PEM
  c. 可复制到剪贴板
  d. 将证书发给需要向您发送加密配置的开发者
- 导入配置包
  a. 点击**「导入配置包」**
  b. 选择 `.tscfg` 或 `.json` 文件，或粘贴 JSON 内容
  c. 点击**「仅验证」**预览，或**「导入」**保存到设备
  d. 导入后配置包写入设备加密存储

- 导出配置包

  前提：仅 **Developer** 设备可用。

  a. 点击**「导出配置包」**
  b. 在文件浏览器中选择配置文件（可多选、选整个目录）
  c. 填写配置名称（必填）、描述（可选）
  d. 粘贴目标设备的证书 PEM（必填，用于加密）
  e. 5. 点击**「生成配置包」**
  f. 生成的 `.tscfg` 可复制或下载，发给目标设备

- 查看配置包列表

  a. 点击**「查看配置包列表」**
  b. 输入或选择目录路径
  c. 点击「**刷新**」，查看该目录下 `.tscfg` 文件列表
  d. 表格显示：文件名、大小、签名者、官方、状态、操作

使用流程示例

- SSH 免密登录

  a. 生成新密钥（如 `default`）
  b. 点击该密钥「**部署**」→ 填写目标主机、用户名、密码 → 开始部署
  c. 主机出现在「**已部署主机**」
  d. 之后使用该主机连接时无需密码

- HTTPS 证书配置

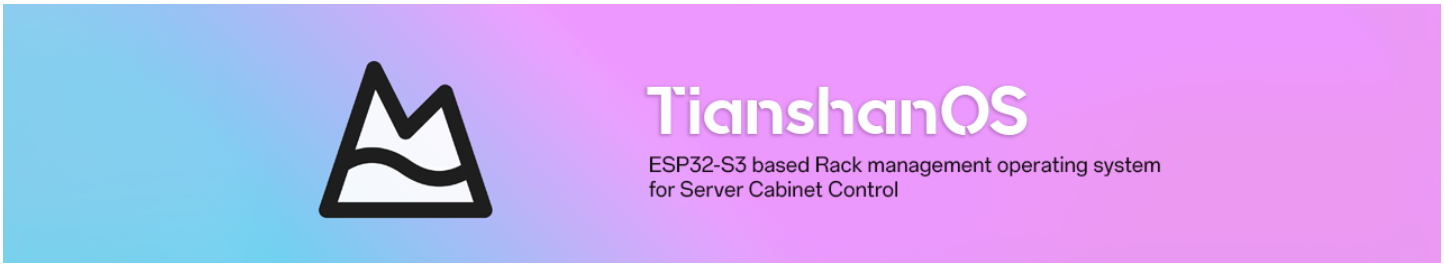
  a. 生成密钥对
  b. 生成 CSR → 在 CA 服务器签发
  c. 安装证书 →（可选）安装 CA 链
  d. 状态变为「**已激活**」

- 配置包分发

  a. 目标设备在「**配置包**」中「**导出设备证书**」，发给开发者
  b. 开发者在 Developer 设备上「**导出配置包**」，选择文件并粘贴目标证书
  c. 目标设备「**导入配置包**」，完成配置接收

> ℹ️ **注意事项**
>
> - 密钥存储在设备 NVS 加密分区，重启不丢失
> - 私钥导出有泄露风险，仅在必要时勾选「**允许导出私钥**」
> - ECDSA 密钥不支持 SSH 公钥认证，请使用 RSA
> - 导出配置包仅 Developer 设备可用
> - 主机指纹不匹配时，务必先核实再选择是否更新

TianshanOS User Getting Started Guide



System

Resource Monitoring

- The resource monitoring page is used to display the system resource status of the **out-of-band management chip** built into the RM-01 fuselage.
- In the modal dialog that appears after clicking the **Service** button, the health status of each running service in the out-of-band management system is displayed.
- In the modal box that pops up when you click the **Details** button, the **memory** and **storage** usage details of the out-of-band management system are displayed, which are used to view the usage and potential issues of each type of memory.
- If you are unsure about the functions of each service, please do not perform any operations; if you have confirmed the service's role and need to handle an exception, you can manually restart the corresponding service.

System Overview

- The System Overview page displays **the operating status of TianshanOS** and **key information about RM-01** .
- On the System Overview page, you can view system information and power status, and perform the following operations: restart **the TianshanOS** system, set shutdown logic (only applicable to models equipped with lithium batteries), or configure the RM-01 top USB interface. Before performing any critical operation, please evaluate and confirm its potential impact.
- The USB button is used to switch the connection target of the top USB interface of RM-01 among **the out-of-band management chip, inference module, and application module** to adapt to different scenarios or peripheral accessories, with the default target being **ESP** . Incorrect operation may cause the current device to disconnect or malfunction. If you are unclear about this function or do not have the corresponding accessories, please consult the dealer or operation and maintenance personnel first, and do not operate it casually.
- When **TianshanOS** experiences incomplete loading, laggy operations, or other anomalies, you can restart the out-of-band management system via the "Restart" button to restore services. The restart process may cause a short interruption; if you are unsure about the scope of impact, please contact the distributor or operations and maintenance personnel first.
- **Power Protection** button is used to enable or disable the low voltage protection strategy: automatically trigger the shutdown and recovery process when the input voltage drops abnormally, to prevent damage or instability caused by long-term operation of the device under low voltage.

Network and Time

- Network and Time centrally displays network connectivity, current IP, device time, synchronization status, and time zone, and provides access to time synchronization, time zone settings, and OTA upgrades, facilitating quick checks and adjustments.
- **Sync Time** button is used to write the current system time of the browser into the out-of-band management system (TianshanOS), which is applicable to scenarios where the device time is inaccurate due to offline/no NTP, RTC battery failure, or when quick time calibration is required.
- **Time Zone** button is used to select a preset or enter a custom POSIX time zone and apply it to the out-of-band management system, which is used to set the UTC offset and daylight saving time rules to correctly display and parse local time.

- Clicking on “**OTA Upgrade**” will open the firmware upgrade page, providing one-stop online and offline upgrade management for system firmware and WebUI.
  - OTA Mode: The browser downloads the firmware from the server and then uploads it to the ESP32 via the API (the device does not need to be connected to the Internet). After the upgrade is completed, it will automatically restart, and the page will attempt to detect the device coming back online.
  - Partition Management
    - Running Partition: Displays the firmware version, storage address, and size of the currently booted partition; you can select "Mark as Valid" to cancel the automatic rollback mechanism.
    - Bootable partition: Refers to another partition that is not currently running. If it already contains bootable firmware, you can click "Roll back to this version" and restart to boot from this partition to restore to that firmware.
    - Switching/rolling back requires a restart to take effect. Marking as valid and rolling back operations are key measures to ensure the stability of upgrades and rapid recovery.
  - Manual Upgrade
    - Upgrade from URL: Enter the firmware address in the input box, and you can check "Include WebUI" to continue flashing `www` after the firmware is flashed; "Skip Verification" can bypass integrity/signature verification (with significant risks). Applicable scenarios: The firmware is hosted on an HTTP/HTTPS server, and RM-01 or the browser must be able to access the URL.
    - Upgrade from SD card: Fill in the firmware path on the SD (TF/MicroSD) card (e.g., `/sdcard/firmware.bin`). When "Include WebUI" is checked, it will also search for and flash `www.bin` in the same directory; suitable for local upgrades in offline or no-internet environments of the device.

> ⚠ **Risks and Precautions**
>
> Checking "Skip verification" will increase the risk of flashing errors or flashing malicious/corrupted firmware; when WebUI is included, it is a two-step flash, which increases the overall time and requires ensuring compatibility between the two versions.

Device Panel

- Overall Layout: The device panel consists of two upper and lower areas and three buttons in the upper right corner. The upper area is for quick operations, the lower area is for data monitoring, and the three buttons are Inference Module Power Control, Application Module Power Control, and Component Management respectively.
- Upper Area
  - Quick actions will be displayed as clickable cards, with each card corresponding to an automation rule.
  - Source and Configuration: The cards come from an automated system (dedicated interface for publishers or resellers), with **manually triggered** rules enabled and checked. Publishers or resellers can pre-configure these rules to provide to users.
  - Usage: Click on the card to execute the corresponding rule. For example: launch an app, execute a script, send a command, etc.
  - Additional capabilities (if the rule includes background execution):
    - Displays the running status of the process (green dot indicates running)
    - Provide " **Log** " and " **Stop** " buttons for viewing logs or terminating processes
    - When the process is running, clicking the card again will prompt "The process is running, please stop it first".
    - Empty status: If there is no rule that meets the condition, "No shortcut yet" will be displayed and a prompt will be given to enable the " **Manual trigger** " option in the automation rule.
- Lower Area
  - The lower area is a user-defined data component for real-time display of monitoring data.
  - Add and Edit: Click on the **Component Management** in the top right corner, and in the pop-up window:
    - On the left, you can set the auto-refresh interval (e.g., 1 second, 5 seconds, 10 seconds, or disable it)
    - New components can be added (from preset or custom types)
    - Edit, move up/down, and delete existing components
  - You can also directly click on a component card on the panel to enter the editing interface of that component.
    - Preset component types: CPU, Memory, Hard Disk, Temperature, GPU, Power Consumption, Voltage, Current, Network Speed, Status, Runtime, Log Stream, etc., used for quickly adding common monitoring items.

    - Custom component types: Circular Progress, Gauge, Thermometer, Number, Progress Bar, Text, Status, Icon, Dual Value, Percentage, Log Stream, etc., can be bound to any automation variable expression (such as `\${cpu.usage}`, `\${power.watts}`).
    - Configuration Saving: Component configurations will be saved to the device (SD card first, NVS as backup), and will still be retained after changing browsers or restarting the device.
    - Empty state: When no components are added, display "No data components have been added yet", guiding users to add them through component management.
- The two green buttons in the upper right corner of the device panel monitor the status of **the inference module** (AGX running) and **the application module** (LPMU running); **green** indicates that the device is running, **yellow** indicates that the device status is being retrieved (usually starting or connecting), **red** indicates that it is turned off (powered off). Users can also click to force the module to start/stop.

> ⚠ **Safety Notice**
>
> - Forced power-off or triggering power switching during module operation may cause irreversible damage or loss of data and content.
> - Before execution, please confirm that the current work has been saved or that you are aware of the possible consequences.

Fan Control

- Overall Layout: Fan control is located on the system overview page, side by side with the device panel. It includes**temperature status bar**,**fan card grid**, and two buttons in the upper right corner:**Refresh**,**Curve**.
- Top Right Button
    - **Refresh** Button: Used to re-acquire the current fan status (mode, speed, RPM, temperature, etc.) to ensure that the display shows the latest data.
    - **Curve** Button: Opens the fan curve management pop-up window, used to configure the temperature - speed curve and related parameters.
- Temperature Status Bar, displaying three types of information horizontally at the top
    - Effective Temperature: The temperature value (°C) that can be used for speed control, sourced from the bound temperature variable; if not bound or invalid, it will be displayed in a warning color

- Target speed: In curve/auto mode, it is the target duty cycle (%) calculated based on the current temperature; in manual mode, it is the actual current speed
- Test Temperature: Manual temperature input for debugging curve logic

> ⚙️ Test temperature function
>
> - Enter an integer between 0 and 100 (unit: °C) in the input box, click "Test" to switch the temperature source to this fixed value, and the fan will calculate the target speed based on this temperature according to the curve.
> - It is suitable for quickly verifying the rotational speed performance at different temperature points when the real temperature change is not significant.
> - Clicking "Clear Test" will resume using the actual temperature variable and no longer use the manual temperature.
>
> **Note** : During the effective period of the test temperature, the fan speed control no longer relies on real sensors, is only used for debugging, and should be cleared as soon as possible after verification.

- Fan Card
  - Card Title and Mode
    - Top left corner: Fan number, e.g., **Fan 0**.
    - Top right corner: Current mode badge
      - **Off**: Fan stops rotating
      - **Manual**: Fixed speed specified by the user
      - **Automatic**: Use the system default speed control strategy
      - **Curve** : Use custom temperature - speed curve
  - Speed Display
    - The large number represents the duty cycle (0–100%).
    - Curve/Automatic Mode: Displays the target speed.
    - Manual Mode: Displays the currently set rotational speed.
    - If the fan supports RPM feedback, the actual rotational speed (RPM) will be displayed below.
  - Mode Switch: Four Tab Buttons: **Off** | **Manual** | **Automatic** | **Curve** , click to switch the operating mode of the fan.

- Speed adjustment slider
  - Only available in manual mode; grayed out and unavailable in off, automatic, and curve modes.
  - Range 0–100%, drag the slider or enter a value to set the fan speed.

> ℹ️ **Hint**: "Adjustable after switching to manual mode" indicates that you need to switch to manual mode first to use the slider.

- The Curve Management Popup is opened by clicking the **Curve** button, and it can configure the temperature - speed curve and its related parameters.
  - Fan Selection: Select the fan to be configured (e.g., 0, 1, etc.) from the top dropdown box.
  - Temperature Source (Bound Variable)
    - Function: The curve mode needs to calculate the rotational speed based on temperature, and the temperature source is the bound automation variable (e.g., `agx.cpu_temp`).
    - Operation: Select a floating-point variable in the variable drop-down box and click " **Bind** "; you can **unbind** or replace the variable at any time.
    - Status: Displays the currently bound variable and its real-time temperature value; when not bound, it will show "Unbound".
    - **Hint**: You can choose a CPU temperature variable similar to `agx.cpu_temp` as the temperature source.
  - Temperature - Speed Curve
    - consists of several (temperature, speed) points, each of which represents "when the temperature reaches X°C, the target speed is Y%".
    - Add Point: Click on **Add Point** to add a new curve point at the end (up to 10 points).
    - Delete Point: The delete button next to each point can remove that point (at least 2 points must be retained).
    - Edit: Modify the temperature (-20–120°C) and rotation speed (0–100%) in the input box.
    - **Rule**: When the temperature is below the minimum point, use the minimum speed; when it is above the maximum point, use the maximum speed; in between, perform linear interpolation according to the curve.
- Curve Preview: The canvas draws the temperature - speed relationship of the current curve in real time, facilitating the inspection of whether the configuration is reasonable.

- Advanced Parameters

Parameter	Meaning	Scope / Description
Minimum Duty Cycle	Lower limit of target rotational speed calculated by the curve	0–100%, usually above 20% can avoid stalling
Maximum Duty Cycle	Target speed upper limit calculated by the curve	0–100%, typically 100% represents full speed
Temperature Hysteresis	Switching threshold difference when temperature rises/falls	0–20°C, used to reduce frequent speed regulation near the critical temperature
Minimum Interval	Minimum time interval between two speed adjustments	500–30000 ms, used to avoid overly frequent adjustment

- Bottom Operation
  - Cancel: Close the pop-up window without saving changes.
  - Import Configuration: Load existing curve configuration from local or SD (TF/MicroSD) card.
  - Export Configuration: Export the current curve configuration to local and SD (TF/MicroSD) card.
  - Apply Curve: Write the current configuration to the device and make it effective.

> ℹ️ **Precautions**
>
> - The curve mode requires binding the temperature variable first; otherwise, it cannot control the speed correctly according to temperature.
> - The minimum duty cycle should not be set to 0, otherwise it may stop rotating for a long time at low temperatures.
> - The test temperature is only used for debugging. After the test is completed, the test must be cleared to resume normal speed control.

LED Control Panel

- Overall Layout: LED control is located on the system overview page, below the fan control, consisting of a title bar button and a grid of device cards, with each physical LED device corresponding to a card.
- Title Bar Button

Button	Function
Refresh	Reacquire the list of all LED devices and their current status
Color Correction	Open the "Global Color Correction" pop-up window, which is only displayed when a Matrix screen device exists
Close All	Turn off all LED devices and clear the display content

- Device Type: The system supports three types of LED devices, returned by the backend `led.list`, with each type corresponding to a card.

Device Name	Description	Typical Applications
Touch	Touch indicator light (1 WS2812)	Status Indication
Board	Motherboard Status LED Strip (28 WS2812)	Status Light Strip
Matrix	LED Matrix Screen (16×16)	Images, text, QR codes, filters, etc.

- Device Card Structure
  - Status Badge: Currently Displayed Status
    - **Closed** : Device is off
    - **Constant Brightness**: Solid color display, no animation
    - **Animation Name**: Playing Program Animation
  - Stop Animation: Clickable when there is an animation currently playing to stop the current animation
- Control Area
  - Brightness Slider
    - Range 0–255
    - Drag to take effect, and it will be immediately sent to the device
    - The current brightness value is displayed on the right
  - Color Selection
    - Custom Color: Click on the color block to open the system color picker, and it will be immediately filled into the device after selection
    - Quick color blocks: red, orange, yellow, green, cyan, blue, purple, white, pink, gray, etc. Click to fill; the filled color will automatically turn on the device and enter the always-on state.
- Bottom Action Bar

Button	Function
Power Supply (Light Bulb Icon)	Switch device: Off → Click once for constant white light; On → Click once to turn off
More Animation (Play Icon)	Open the "Program Animation" pop-up window, select and start the animation
Image/QR Code	Only Matrix display, open image/QR code pop-up window
Text	Only Matrix display, open text display pop-up window
Filter Effect	Only Matrix display, open the post-processing filter pop-up window after opening
Save	Save the current display state as the boot configuration and restore it after restarting

- The program animation pop-up window is opened by clicking **More Animations** , and you can switch and configure animation effects.
  - Enable Switch: Checking/Unchecking is equivalent to the power switch
  - Animation List: Button Grid, click to select and preview, some animations support color parameters (e.g., breathing, solid, rain)
  - Setting: Display when selected
  - Speed: 1–100 slider, controls animation speed
  - Color: Only partially animated, used to select the primary color
  - Operations: Reset, Apply (start the selected animation and close the pop-up window)
  - **Description**: The animation types are provided by the backend `led.effect.list` , and may vary for different devices.
- Image / QR Code Popup (Matrix Only)
  - Image:
    - Path Input Box: Enter the image path (e.g., `/sdcard/images/xxx.png`)
    - Browse: Open the file picker and select an image from the SD card
    - Center Display: Check to center the image
    - Display Image: Display the image at the current path to the Matrix
  - QR Code:
    - Content Input: Enter text or URL to generate QR code
    - Error correction level: L / M / Q / H
    - Foreground color: QR code color
    - Background Image: Optional background image path

Generate QR Code: Generate and display it on the Matrix

- Text display pop-up (Matrix only)
  - Text content: Enter the string to be displayed
  - Font: Select from the system / SD card font list
  - Alignment: Left / Center / Right
  - Color: Text Color
  - Position: X, Y coordinates, or check "Auto Position"
  - Scroll: None / Left / Right / Up / Down, as well as speed and whether to loop
  - Display: Apply and display text
  - Stop: Stop text display
- Post-processing filter pop-up window (Matrix only): Overlay real-time filter effects on existing content.
  - Filter list: Pulse, Breath, Blink, Wave, Scanline, Glitch Art, Rainbow, Shine, Plasma, Nostalgia, Color Separation, Contrast, Invert, Canary Release, etc.
  - Operation process: Select filter → Optional adjustment parameters (speed, intensity, wavelength, etc.) → Click " **Apply** " → Click " **Stop** " to stop the filter.
- Global Color Correction pop-up: The **Color Correction** button appears in the title bar only when the Matrix device is present. It is used to correct the display color of the Matrix.

Parameter	Function	Scope
White Balance	Adjust the R/G/B channels separately to correct the color temperature	Each channel 0–2.0, default 1.0
Gamma	Midtone Contrast	0.1–4.0, default 1.0
Brightness	Overall Brightness	0–2.0, default 1.0
Saturation	Color Saturation	0–2.0, where 0 is the canary release

ℹ️ Precautions

- Saving the configuration will write the currently displayed content to the boot configuration, which will automatically restore after a restart.
- Color correction is applied to the global output of Matrix, and adjustments will affect all displayed content.
- If no LED device is detected, please check the LED service status and GPIO configuration.

Network

Overall layout

The web page is accessed through the sidebar " **Network** " and consists of three main areas: network status overview, interface configuration, and network services, as well as several expandable auxiliary panels and pop-ups.

Network Status Overview

The horizontal row at the top of the page displays the brief status of the current three network interfaces:

Interface	Display Content	State Meaning
Ethernet	Link Status (Connected/Disconnected), IP Address	Green = Connected, Gray = Not Connected
WiFi Client	Connection Status (Connected/ Disconnected), IP Address	Green = Connected, Gray = Not Connected
WiFi Hotspot	Operating Status (Running/Not Enabled), Number of Connected Devices	Green = Running, Gray = Not Enabled

Interface Configuration Panel

Switch between two tabs: **Ethernet** and **WiFi** .

- Ethernet, showing the read-only status of the Ethernet interface without providing a modification entry:

item	Meaning
Link Status	Connected / Not Connected
IP Address	Local IP
Subnet Mask	Subnet Mask
Gateway	Default Gateway
DNS	DNS Server
MAC Address	Network card physical address

> ℹ️ **Explanation**: Ethernet addresses are typically assigned by the superior network (such as DHCP) or configured through other means.

- WiFi
  - WiFi Mode Selection:
    - Drop-down selection: `Off / Station (STA) / Access Point (AP) / STA+AP`
    - Switch mode and refresh the page immediately after selection
  - Station Connection Area (displayed in STA or APSTA mode):

Item	Meaning
Status	Connected / Not Connected
SSID	Currently connected WiFi name
IP	Local Machine IP
Signal	Signal Strength dBm and Brief Indicator Bar

  - Operation Button
    - Scan: Open the WiFi scan panel to list available networks
    - Disconnect: Disconnect from the current WiFi (only displayed when connected)
  - Hotspot Area (displayed in AP or APSTA mode):

item	Meaning
Status	Running / Not Enabled
SSID	Hotspot Name
IP	Hotspot IP (e.g., 192.168.4.1)
Number of connections	Number of currently connected devices

- Operation Button:
  - Configuration: Open the hotspot configuration pop-up window
  - Device: Open the list of devices connected to the hotspot

---

WiFi Scan Panel

Click " **Scan** " and expand to display a list of available WiFi.

- Each network displays: SSID, signal strength dBm, channel, authentication method
- Click on a network to pop up the password input box, enter the password and then initiate the connection (leave it blank for open networks)
- Refresh: Rescan
- Close: Collapse the panel

> ℹ️ **Explanation**: Scanning can only be performed in STA or APSTA mode; if the mode is off or ap, a prompt saying "Need to switch to STA or APSTA mode" will appear.

Hotspot Access Device Panel

Click " **Devices** " and expand to list the devices currently connected to the hotspot.

- Each device displays: MAC address, signal strength in dBm
- Close: Collapse the panel

DHCP Client Panel

After clicking "Client", it expands to display the DHCP allocation status.

- Interface selection: Select " **WiFi AP** " or " **Ethernet** "
- The list shows: IP, MAC, Hostname (if available), Type (Static/Dynamic)
- Refresh: Reacquire the client list
- Close: Collapse the panel

Network Service Panel

- Hostname
  - Current hostname (e.g., TianshanOS)
  - Enter the new hostname → Click "Settings" → Update the hostname and refresh the page
- DHCP Server
  - Status Badge: Displays the number of running DHCP interfaces (e.g., 2/2) or the overall status
  - Interface List: Operating Status and Lease Count of Each Interface
  - Client: Open the DHCP Client panel to view the allocation status
- NAT Gateway
  - Status Badge: Running / Stopped
  - Interface Status: Whether WiFi and Ethernet are available (for NAT)
  - Enable/Disable: Toggle the NAT switch
  - Save: Save NAT configuration

> ℹ️ **Description**: NAT is used to share internet access between WiFi and Ethernet. Enabling requires simultaneously meeting the following conditions:
>
> - WiFi is connected, and
> - Ethernet is connected
>
> Otherwise, the button will be disabled.

WiFi Hotspot Configuration Popup

Click "Configuration" to open, which is used to set the hotspot parameters in AP or APSTA mode.

Z	Meaning	Description
SSID	Hotspot Name	Required
Password	Wireless Password	Leave it blank for an open network; if filled in, it must be at least 8 characters long
Channel	Working Channel	2001/6/11
Hide SSID	Whether to hide the hotspot name	After checking, you need to manually enter the SSID to connect

---

Example of Usage Process

- Connect WiFi: Select STA or APSTA in WiFi Tab → Click " **Scan** " → Select Network and Enter Password → Connect
- Disconnect WiFi: Click <**Disconnect** when connected
- Enable hotspot: Select AP or APSTA → Click " **Configure** " → Fill in SSID, password, etc. → Apply
- View hotspot access devices: Click " **Devices** " → Expand the list of access devices
- Enable NAT: Ensure both WiFi and Ethernet are connected → Click <**Enable**
- Modify Hostname: Enter the new hostname in the service area → Click <**Settings**
- View DHCP Client: Click " **Client** " → Select interface → View allocation list

---

ℹ️ Precautions

- Ethernet is configured for read-only display, and modifications must be made through the superior DHCP or other means
- Scanning and connecting to WiFi requires STA or APSTA mode
- Hotspot password must be at least 8 characters long, or leave it blank to set it as an open hotspot
- Before enabling NAT, please confirm that both WiFi and Ethernet are properly connected

---

File

Overall layout

The File Page is used to browse and manage **files and directories stored on the Out-of-Band Management System (TianshanOS)** . It mainly includes **Toolbar** , **Storage Switch** , **Navigation** , **File List** , **Uninstall** and other areas and buttons.

Toolbar

Button	Function
Upload File	Open the upload pop-up window and upload local files to the current directory
New Folder	Open the new folder pop-up window and create a subfolder in the current directory
Refresh	Reload the current directory list

Storage Switching

Two labels are used to switch storage locations:

Label	Path	Description
SD Card	/sdcard	External SD Card Storage
SPIFFS	/spiffs	On-board Flash storage

Clicking on the corresponding label will jump to the root directory of the storage.

SD Card Control:

- Mounted: Displays the "Unmount SD" button, allowing the SD card to be unmounted
- Unmounted: Displays the "Mount SD" button, allowing the SD card to be mounted
- When the current path is the SD card path and the SD card is not mounted, the list area will display the "SD card not mounted" message and the "Mount SD card" button

Storage Status: The bottom displays the mounting status (mounted / unmounted) of SD and SPIFFS.

File List

Display the contents of the current directory in tabular form, with columns including:

Column	Description
Checkbox	Select/Unselect this item, and the table header can be used to "Select All"
Name	File or folder name; click on the folder name to enter the directory
Size	File size (B/KB/MB/GB), directories are displayed as -
Operation	Download (files only), rename, delete

Batch Operation Toolbar

After checking at least one item, the following will appear above the file list:

Button	Function
Batch Download	Download the selected files (excluding selected folders)
Batch Delete	Delete selected files and folders (confirmation required)
Unselect	Clear the current selection

Tip: When downloading in bulk, folders will be automatically ignored; if only folders are selected, a prompt saying "There are no downloadable files among the selected items" will appear.

Upload

Opens after clicking "Upload File". Usage:

- Click on the upload area to select a file, or drag and drop a file onto the upload area
- Supports multiple selection
- Selected files will be listed in the upload list and can be removed one by one

ℹ️

- Target Path: Upload to the currently browsed directory, with the file name unchanged.
- Upload limitation: The backend only accepts writing to /sdcard, and uploading to SPIFFS will fail.
- Configuration Package (.tscfg): When uploading a.tscfg file, the signature will be automatically verified, and after verification, you can choose whether to apply it immediately.

ℹ️ Precautions

- Upload path: Only uploading to /sdcard is supported, and writing to /spiffs is not allowed.
- Deletion is irreversible: Once deleted, it cannot be undone. Please confirm before deleting.
- SD Card Unmount: After unmounting, files on the SD card cannot be accessed; if currently in the SD path, it will automatically switch to /spiffs after unmounting.
- Batch Download: Only selected files will be downloaded, and selected folders will be automatically ignored.

Safety

ℹ️

- This chapter is the " **security** " page introduction and operation summary, detailed operation steps refer to the **安全页面完整操作指南（admin 用户）/ Complete Operation Guide for Security Page (admin user)**
- The Security Page is the security management center of TianshanOS, centrally managing SSH keys, remote hosts, HTTPS certificates, and configuration packages. It requires **admin** privileges, and the URL route is `#/security`.

Overview and Preparation

- How to access the security page

a. Access the device's WebUI in the browser (e.g., `http://192.168.4.1`)
b. Log in using the admin account
c. Click on the **"Security"** link in the top navigation bar
d. Go to the security page (`#/security`)

- Page Structure

  The page is divided into five sections from top to bottom:

Block	Function
Key Management	Generate, export, deploy, revoke, and delete SSH keys
Deployed Host	List and operations of remote servers with deployed public keys
Known host fingerprint	SSH Server Fingerprint, Protect Against Man-in-the-Middle Attacks
HTTPS Certificate	Device mTLS Key and Certificate Management
Config Pack	Device Certificate Export, Configuration Package Import/Export, and List

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Key Management

- Generate SSH Key
  - Operation Steps
    i. Click on the **"Generate New Key"** button in the "Key Management" block
    ii. Fill out the form in the pop-up window
  - Form Field

Field	Required	Description	Fill-in Example
Key ID	is	Unique identifier, consisting of letters, numbers, and underscores	default、agx_ke
Key Type	is	Encryption Algorithm	RSA 2048-bit（推荐）、RSA 4096-bit、EC
Remarks	No	Usage Instructions	Used to connect AGX
Alias	No	Alternative ID display, recommended to fill in when "Hide Key" is enabled	AGX Production Envirc
Allows exporting private keys	No	Check to export private key backup	Check only when nec
Hide Key ID	No	Low-privilege users cannot see the real ID	Check when multiple people s

- Key Type Description

Type	Generation Time	Attention
RSA 2048-bit	5-10 seconds	Recommended, SSH public key authentication support
RSA 4096-bit	30-60 seconds	High security, SSH support
ECDSA P-256	3-5 seconds	SSH public key authentication is currently not supported
ECDSA P-384	3-5 seconds	SSH public key authentication is currently not supported

i. Click **the "Generate"** button and wait for completion (RSA 4096 takes approximately 30-60 seconds, do not close the page)

ii. After the pop-up window prompts success, it closes, and the new key appears in the table

> ℹ️ **Precautions** :
>
> - Key ID cannot be duplicated; an error will be reported if it already exists
> - Do not refresh or close the page during the generation process
> - The key is stored in the NVS encrypted partition and will not be lost upon restart

- Deploy the public key to the remote server
  - Prerequisites
    - At least one SSH key has been generated

- Know the IP/hostname, username, and password of the remote server
- Device is network-connected to the server

- Operation Steps

  i. Find the key to be deployed in the **Key Management** table

  ii. Click the **"Deploy"** button in this row

  iii. Fill in the pop-up window titled "Deploy Public Key to Remote Server":

Field	Required	Description	Example
Target Host	is	IP or hostname	192.168.55.100, agx.local
Username	is	Remote Login User	root, nvidia
Port	No	SSH port, default 22	22, 2222
Authentication Password	is	SSH login password, required for the first deployment	—

  iv. Click **"Start Deployment"**

  v. The pop-up window displays the deployment process: Connect → Verify host key (trust required for the first time) → Upload public key → Verify

  vi. After success, the main host will automatically be added to the "Deployed Hosts" list

- Common Errors

Error	Possible Reasons	Process
Connection timed out	Network connectivity issues, firewall, SSH not started	Check the network, confirm that sshd is running, and check the firewall
Password is incorrect	Username or password is incorrect	Check username, password, and case sensitivity
Permission Denied	SSH does not allow password login	Enable PasswordAuthentication yes in the server sshd_config
Host fingerprint verification failed	Trust required for the first connection	See "Known Host Fingerprint"

- Revoke Public Key

  - Revoked from Key Management

    i. Find the key to be revoked in the **Key Management** table

    ii. Click **the "Undo"** button

iii. Fill in the following in the pop-up window: target host, username, port (default 22), authentication password

iv. Click **"Revoke Public Key"** , and after confirmation, remove the public key from the remote `authorized_keys` , and the local record will be deleted accordingly

- Revoke from deployed hosts

  i. Find the target host in the **Deployed Hosts** table

  ii. 2. Click the **"Undo"** button

  iii. Enter the server password and confirm

  iv. The system deletes the remote public key and removes the local record

- Remove only local records

  > If the server has been deactivated or the public key has been manually deleted, simply clean up the local environment:

  i. 1. Click **"Remove"** in **"Deployed Hosts"**

  ii. 2. After confirmation, only local records will be deleted, without modifying the remote `authorized_keys`

- Export Key

  - Export Public Key

    i. Find the key in the **Key Management** table

    ii. Click **the 「Pub」** button

    iii. Copy the public key or download it in the pop-up window `.pub` file

    iv. You can paste the public key into the remote `~/.ssh/authorized_keys`

  - Export Private Key

    > **Premise** : " **Allow export of private key** " is checked when generating the key.

    i. Click **the 「Priv」** button

    ii. Read the safety warning and confirm

    iv. Copy the private key or download the `.pem` file

    ℹ️ **Caution**: Private key leakage can lead to unauthorized access. Do not send it via email or chat

- Delete Key

  a. Find the key to be deleted in "Key Management"

b. Click **the "Delete"** button
c. Confirm deletion (**irreversible**)
d. The key has been permanently deleted from NVS; the deployed host records remain and need to be manually revoked or removed

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Deployed Host

- Import Host Configuration

  > Applicable to: The public key is deployed via the terminal or manually, and the host does not automatically appear in the list.

  a. In the "Deployed Hosts" section, click **"Import Hosts"**
  b. Select `.tscfg` configuration package file
  c. The system verifies the signature and displays the configuration ID, signer, etc.
  d. Check "Overwrite existing configuration" (optional)
  e. Click **"Confirm Import"**
  f. Refresh after import, and the host will appear in the list
- Export Host Configuration
  a. Find the host in the "Deployed Hosts" table
  b. 2. Click the **「Export」** button
  c. Optional: Paste the target device certificate PEM, used for distribution to other devices
  d. Click "Export" to download the `.tscfg` file
- Test SSH Connection
  a. Find the target host in "Deployed Hosts"
  b. 2. Click the ****「Test」**** button
  c. The system uses the deployed key to connect and execute simple commands
  d. Check Results: Display username on success, display error message on failure
- Host List Field

Column	Description
Host ID	Format: Key ID@Host Address
Address	IP or hostname
Port	SSH Port
Username	Logged-in User
Deployment Key	Key ID used

Known host fingerprint

> Server fingerprints automatically recorded during SSH connections, used to prevent man-in-the-middle attacks.

- First connection to a new server

  When connecting for the first time, the system will prompt you to trust the host:

  a. Displays the host address, key type, and fingerprint (SHA256)

  b. Select**yes**: Trust and save fingerprint, continue connecting

  c. Select**no**: Abort connection, do not save

  After trust, the fingerprint appears in the " **Known Host Fingerprint** " list, and subsequent connections are automatically verified.

- Host fingerprint mismatch

  If the server fingerprint does not match the saved one, a red warning will pop up.

  Possible causes : man-in-the-middle attack, server reinstallation, key replacement, IP reallocation.

- View and Delete Fingerprint
  - View full fingerprint

    i. Find the host in the "Known Host Fingerprint" table

    ii. 2. Click the 「**View**」 button

    iii. Popup displays the full SHA256 fingerprint
  - Delete Fingerprint

    i. Find the host to be deleted in the table

    ii. Click 「**Delete**」

iii. Confirm deletion

iv. The next time you connect to this server, you will be prompted to trust the fingerprint again

> ℹ️ Tip: Before reinstalling the server, you need to delete the old fingerprint first; otherwise, it will continuously prompt a mismatch.

HTTPS Certificate

- Certificate Status Explanation

Status	Description
Key not generated	Key pair needs to be generated first
Key generated, certificate not installed	CSR needs to be generated and the CA-issued certificate installed
Activated	The certificate has been installed and is valid
is about to expire	The certificate will expire within 30 days, it is recommended to renew it
Expired	Certificate expired, mTLS unavailable

- Complete Certificate Application Process
  - Step 1: Generate a key pair
    i. Click **"Generate Key Pair"**
    ii. Click directly in the pop-up window **「Generate」** (ECDSA P-256, generated on the device side, not exported)
    iii. If a key already exists, a prompt to overwrite will appear, and after confirmation, the process will continue
  - Step 2: Generate CSR
    i. Click **「Generate CSR」**
    ii. Fill in (optional) in the pop-up window:

Field	Description	Example
Device ID (CN)	Device Unique Identifier	TIANSHAN-RM01-0001
Organization (O)	Organization Name	HiddenPeak Labs
Organizational Unit (OU)	Department	Device

iii. Click 「**Generate CSR**」

iv. Copy the CSR content and submit it to the CA server for issuance

- Step 3: Install the device certificate

  i. Issue a certificate using CSR on the CA server

  ii. Click **"Install Certificate"**

  iii. Paste the issued PEM certificate into the text field

  iv. Click 「**Install**」

- Step 4: Install the CA Certificate Chain (Optional)

  i. 1. Click 「**Install CA**」

  ii. Paste the root certificate and intermediate certificates (in PEM format, multiple can be concatenated)

  iii. 3. Click 「**Install**」

- Step 5: View Certificate

  i. Click 「**View Certificate**」

  ii. Popup window displays the certificate PEM, which can be copied to the clipboard

- Delete Voucher

  a. Click 「**Delete Voucher**」

  b. Delete the private key, device certificate, and CA chain after confirmation

  c. The operation cannot be undone

Config Pack

Used for secure encryption and signature configuration, distributed among devices.

- Export device certificate

  a. Click **"Export Device Certificate"**

   b. Pop-up window displays: Certificate Fingerprint (SHA256), Certificate CN, Certificate PEM
   c. Can be copied to clipboard
   d. Send the certificate to developers who need to send you encrypted configurations
- Import configuration package
   a. Click **"Import Configuration Package"**
   b. Select `.tscfg` or `.json` file, or paste JSON content
   c. Click **"Verify Only"** to preview, or **"Import"** to save to device
   d. After import, the configuration package is written to the device's encrypted storage
- Export Configuration Package

   Prerequisite: Only **Developer** devices are available.

   a. Click **"Export Configuration Package"**
   b. Select configuration files in the file explorer (multiple selection and selecting an entire directory are supported)
   c. Fill in the configuration name (required) and description (optional)
   d. Paste the certificate PEM of the target device (required, for encryption)
   e. 5. Click **"Generate Configuration Package"**
   f. The generated `.tscfg` can be copied or downloaded and sent to the target device
- View the list of configuration packages
   a. Click **"View Configuration Package List"**
   b. Enter or select the directory path
   c. Click on **Refresh** to view the list of `.tscfg` files under this directory
   d. The table shows: File Name, Size, Signer, Official, Status, Operation

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Example of Usage Process

- SSH Password-Free Login
   a. Generate a new key (e.g., `default`)
   b. Click the key " **Deploy** " → fill in the target host, username, password → start deployment
   c. Host appears in " **Deployed Host** "
   d. No password is required when using this host for connection thereafter
- HTTPS Certificate Configuration

a. Generate Key Pair
b. Generate CSR → Sign on CA Server
c. Install Certificate → (Optional) Install CA Chain
d. Status changes to " **Activated** "

- Configuration Package Distribution
  a. The target device **exports device certificate** in the " **configuration package** " and sends it to the developer
  b. Developer " **Export Configuration Package** " on the Developer device, select the file and paste the target certificate
  c. Target device " **import configuration package** ", complete configuration reception

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ℹ️ **Precautions**

- The key is stored in the encrypted partition of the device's NVS and will not be lost upon restart
- Private key export has a risk of leakage, only check " **Allow private key export** " when necessary
- ECDSA keys do not support SSH public key authentication, please use RSA
- Exported configuration packages are only available on Developer devices
- When the host fingerprint does not match, be sure to verify first before choosing whether to update